



The effect of ambient fine particulate matter (PM_{2.5}) on respiratory diseases in China: a systematic review and meta-analysis

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Published online: 11 March 2020

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Abstract

Only a few recent systematic reviews and meta-analysis studies have quantitatively assessed the effect of short-term exposure to ambient fine particulate matter with an aerodynamic diameter of less than 2.5 µm (PM_{2.5}) on respiratory disease (RD) mortality and RD hospital outpatient visits in the Chinese population. A total of 46 articles were selected from Web of Science, PubMed, China National Knowledge Infrastructure and Wanfang databases. Generic inverse variance method was employed to pool individual effect estimates. Pooled estimates represented the increased risk of daily RD mortality and RD outpatient visits per 10 µg/m³ increase in PM_{2.5} concentrations. Positive associations were identified between short-term PM_{2.5} exposure and RD in China. The risk of RD mortality and RD outpatient visits increased by 0.48% (95% CI 0.40%, 0.55%) and 0.90% (95% CI 0.62%, 1.18%), respectively, per 10 µg/m³ increase in PM_{2.5} concentrations. A low increased risk of RD outpatient visits (0.62%, 95% CI 0.57%, 0.66%) was identified at a high level of annual mean PM_{2.5} concentrations (41.36–110.80 µg/m³) with 1.82% (95% CI 1.72%, 1.92%) at a low level of annual mean PM_{2.5} concentrations (29.86–40.20 µg/m³) through subgroup analysis. Short-term exposure to PM_{2.5} was associated with increased risks of RD mortality and RD outpatient visits. The increased risk of RD outpatient visits was pronounced at a low level of annual mean PM_{2.5} concentrations, but the exact mechanisms of this phenomenon require further investigation. Evidence demonstrated in this study could appeal to Chinese policymakers to improve air quality and alleviate the adverse health impact of PM_{2.5} pollutions.

Keywords PM_{2.5} · Respiratory disease · Mortality · Hospital outpatient visits · Meta-analysis · China

Abbreviations

WHO	World Health Organization
AQG	Air quality guideline
RD	Respiratory disease
PM _{2.5}	Fine particular matter with aerodynamic diameter less than 2.5 µm
TSP	Total suspended particulate

PM ₁₀	Inhalable particulate matter with diameter less than 10 µm
PM _{2.5–10}	Coarse particulate matter with aerodynamic diameter between 2.5 and 10 µm
SO ₂	Sulfur dioxide
NO _x	Nitrogen oxides
RR	Relative risk
OR	Odds ratio

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ER	Excess risk
CI	Confidence interval
LCI	Lower confidence interval
UCI	Upper confidence interval
DOW	Day of week
ICD	International classification of diseases
CNKI	China national knowledge infrastructure

1 Introduction

As the largest developing country, China has been facing some of the worst air quality in recent decades mainly owing to its rapid economic development (Li et al. 2015a). For example, in January 2013, a hazardous dense haze covered 1.4 million km² of China and affected more than 800 million people, subsequently causing unprecedented public concern about the health impacts of air pollution (Li et al. 2015b). A similar case happened in 2016 wherein 338 cities in China suffered from 2464 heavy-polluted hazy days, of which 80.3% was caused by PM_{2.5} (Ministry of Ecology and Environment of the People's Republic of China 2016). In contrast to the declining trends of coal-combustion related pollution, which is dominated by pollutants, such as sulfur dioxide (SO₂), nitrogen oxides (NO_x), and particulate matter with diameter less than 10 µm (PM₁₀), the ambient concentration of PM_{2.5} has been rising and has become one of the most serious environmental problems affecting human health and social development in China (Fang et al. 2016; Ye et al. 2019). Data demonstrated that in some mega-cities, such as Beijing, Shijiazhuang, Chengdu, and Xi'an, the recently reported annual levels of PM_{2.5} were at approximately 87, 144, 89, and 97 µg/m³, respectively, from 2013 to 2014 (Zhao et al. 2017a); the Yangtze River Delta including 16 cities, situated in China's eastern coastal region, with its rapid urbanization and industrialization, reported that the annual average PM_{2.5} concentration was 47.04 µg/m³ in 2016 (Yang et al. 2018); these levels far exceeded the upper limit (10 µg/m³ for annual mean and 25 µg/m³ for daily mean) recommended by the World Health Organization (WHO) Air Quality Guideline (AQG) (2006a).

Among the various air pollutants, which are an intricate mixture of solid particles and gaseous pollutants, particulate matter shows the strongest evidence for adverse health effects (Chen et al. 2012). Evidence has demonstrated that, compared with other particulate pollutants, such as the total suspended particulate (TSP), PM₁₀, and coarse particulate matter with aerodynamic diameter between 2.5 and 10 µm (PM_{2.5–10}), PM_{2.5} tends to pose the greatest health risks on acute and chronic illnesses and premature deaths, predominantly from respiratory outcomes (Dai and Song

2001; Ren et al. 2017). Liu et al. (2019) showed a pronounced increased risk of daily respiratory mortality (0.74%) owing to per 10 µg/m³ PM_{2.5} concentration increase compared with PM₁₀ (0.47%) in their worldwide multicity article. Pathological studies have explained that PM_{2.5} pollutant has a large specific surface area and can float in the atmosphere for a longer time, carrying a large number of harmful substances, such as heavy metals and toxic organic compounds. Compared with other particulates, PM_{2.5} pollutant can more easily enter the human body through the respiratory tract, deeply depositing in the bronchioles and alveoli (Guo and Xiao 2017; Zhang et al. 2016). PM_{2.5} pollutant induces alveolar inflammation by acting on alveolar macrophages and type II alveolar epithelial cells, leading to a decrease in local immunity of the respiratory tract and damaging the normal physiological functions of the lungs (Li and Zhao 2013). Given that PM_{2.5} can exert greater toxicity than larger particles (Cifuentes et al. 2000), the WHO (2006b) recommended using PM_{2.5} rather than PM₁₀ concentrations as indicators of air quality. An experimental study showed that the exposure to 288 µg/m³ of fine urban particles (PM_{2.5}) for 6 h a day for 3 days resulted in 37% mortality in bronchitic rats compared with 0% in control rats (Godleski et al. 1996). Studies have confirmed that PM_{2.5} has gradually become the most harmful particulate pollutant to human health (Liao et al. 2011; Wang et al. 2012; Zanobetti and Schwartz 2009).

In China, PM_{2.5} was included in the Ambient Air Quality Standard in 2012 and the National Air Reporting System in 2013. Since January 1, 2013, 74 cities in the Beijing–Tianjin–Hebei Economic Zone, Yangtze River Delta, and Pearl River Delta began to release real-time air quality information, including PM_{2.5} pollutant, to the society. The number of cities in China that monitor PM_{2.5} has soared to 161 by January 1, 2014. In 2015, the PM_{2.5} reporting system covered all 338 prefecture-level and above cities on a national scale, excluding Hong Kong, Macao, and Taiwan (Ministry of Ecology and Environment of the People's Republic of China 2015). Since then, more epidemiological articles have been published in recent years; they document the quantitative association between short-term PM_{2.5} exposure and the increased risk of RD owing to a large number of high-quality PM_{2.5} pollutant data accessible from the national reporting system, which enabled researchers to have more opportunities to explore this relationship in the Chinese population (Li et al. 2018b). For example, for RD mortality, Lin et al. (2016a) research in the Pearl River Delta of China and Chen's (2013) study in 17 typical Chinese cities have revealed positive associations between PM_{2.5} concentration and RD mortality in the Chinese population. Lin et al. (2016c) study in Wang et al. (2019) research in 13 cities of Jiangsu Province,

China also drew consistent conclusions. For RD hospital outpatient visits, Lu et al. (2017) identified that short-term $PM_{2.5}$ exposure contributed to the increased risk of RD hospital outpatient visits in Beijing. Such similar positive concentration–response associations were also illustrated in Yang et al. (2017) and Duan et al. (2015) epidemiological research.

Given a large number of published epidemiological studies available, employing a systematic review and meta-analysis to integrate the existing evidence on the association between $PM_{2.5}$ and RD in the Chinese population is worthwhile. However, only a few systematic reviews and meta-analysis studies focus on this relationship. Only eight relevant meta-analysis studies were identified after a systematic database research. One study (Newell et al. 2017) was conducted across 12 low-income and middle-income countries (LMICs), which included several Chinese cities, but it was still inadequate to extrapolate totally the exposure–response functions to China owing to the differences of the source, characteristics, and spatial variability of $PM_{2.5}$ and the demographic and healthcare characteristics between China and other LMICs. Six studies (Liu et al. 2015; Lu et al. 2015; Ren et al. 2017; Shang et al. 2013; Xie et al. 2009; Zhong et al. 2017) examined the epidemiological evidence of $PM_{2.5}$ and RD mortality, and only one meta-analysis (Liang et al. 2016) assessed the association between $PM_{2.5}$ concentrations and the RD hospital outpatient visits in China. All these studies have synthesized the epidemiological evidence in or prior to 2017. Since 2017, the pace of epidemiological studies published on the quantitative association between $PM_{2.5}$ and RD has accelerated (as shown in Tables 1, 2). However, no recent systematic review and meta-analysis have been carried out to analyze them. Therefore, the aim of the present study is to systematically collect and summarize the epidemiological evidence to date on RD mortality and RD hospital outpatient visits related to ambient $PM_{2.5}$ air pollution observed in the Chinese population and published in either the English or Chinese peer-reviewed literature to provide more up-to-date evidence to help improve air quality and population's level of health.

2 Methods

2.1 Databases

We systematically searched in PubMed, Web of Science, China National Knowledge Infrastructure (CNKI), and Wanfang database (last entry on 10 April 2019) for epidemiological literature on RD mortality and RD hospital outpatient visits owing to the short-term exposure to ambient $PM_{2.5}$ using the following terms: (1) particle,

particulate matter, fine particulate, PM, $PM_{2.5}$, air pollution, and pollutant; (2) respiratory disease, RD, disease, health, adverse effect, mortality, death, morbidity, hospital, and outpatient visits; and (3) China, Chinese, Pearl River Delta, and Hong Kong. The bibliographies of the articles identified using the search criteria above were also manually screened to expand our review. Moreover, “respiratory disease (RD)” in the present study referred to “all diseases of the respiratory system according to International Classification of Diseases ICD-9 460-519 or ICD-10 J00-J98”, excluding its single or sub-classification. EndNote Software (VersionX7, Thomson Corporation, Stamford, CT, USA) was used to manage the references.

2.2 Literature selection and data extraction

All retrieved literature was included in the present study by using manual selection according to the following inclusion criteria: (1) All Chinese and English epidemiological studies with health outcomes related to RD mortality and RD hospital outpatient visits owing to the short-term exposure to ambient $PM_{2.5}$ in China were included. (2) Studies focusing on $PM_{2.5}$ short-term exposures, defined with a duration of up to 7 days to ambient $PM_{2.5}$ associated with RD mortality and RD hospital outpatient visits, were included. (3) Studies that have quantitative exposure–response associations between RD mortality or RD hospital outpatient visits and $PM_{2.5}$ concentration (relative risk [RR], odds ratios [OR], and excess risk [ER], also called attributable risk [AR], which is defined as the difference between the proportion of subjects in a population with a particular disease who were exposed to a specified risk factor and the proportion of subjects with that same disease who were not exposed) were included (Wacholder et al. 1994); their confidence intervals were at 95%, and individual health risk estimates were expressed according to unit change in $PM_{2.5}$ mass concentration. (4) Subjects in the studies were an all-age and all-gender healthy population. (5) They were exposed to ambient $PM_{2.5}$ in the natural environment. (6) The health risk estimates were measured in the non-specific situations (e.g., occupational, thermal, and seasonal). (7) Respiratory diseases were defined according to ICD-9 460-519 or ICD-10 J00-J98. (8) Study designs of time-series analysis or case-crossover analysis were used. (9) Single-pollutant model results in a specific Chinese city were available. (10) For duplicated or derived articles conducted in the same location, study period, data source, pollutants, or health outcomes, only the most recent publication was included. The study followed the PRISMA guidelines for conducting systematic reviews and meta-analyses (Moher et al. 2009).

We excluded the following criteria: (1) studies conducted in other countries or presenting association between

Table 1 RRs¹ or ORs² of RD³ mortality due to short-term PM_{2.5}⁴ exposures

Articles	Study period	Study design	City	Annual mean concentration of PM _{2.5} (µg/m ³)	RR or OR and 95% CI ⁵			Adjusted confounding factors
					RR	LCI ⁶	UCI ⁷	
Li et al. (2015c)	2005–2009	Time-series	Beijing	82.02	1.0036	0.9941	1.0130	Long-term and seasonal trends, temperature, relative humidity, DOW ⁸
Li et al. (2018c)	2009–2012	Case-crossover	Beijing	84.90	1.0043	1.0005	1.0081	Long-term and seasonal trends, temperature, relative humidity, DOW
Li et al. (2013b)	2005–2009	Time-series	Beijing	75.00	1.0063	1.0028	1.0083	Long-term and seasonal trends, temperature, relative humidity, DOW, public holiday
Li et al. (2013a)	2004–2009	Time-series	Beijing	76.00	1.0051	1.0031	1.0066	Long-term and seasonal trends, temperature, relative humidity, DOW
Cui et al. (2018)	2013–2015	Time-series	Jinan	95.00	1.0052	1.0009	1.0096	Long-term and seasonal trends, temperature, relative humidity, DOW, air pressure, wind velocity, public holiday, warm-heating
Wang (2017)	2014–2015	Time-series	Lanzhou	53.18	1.0096	1.0045	1.0147	Long-term and seasonal trends, temperature, relative humidity, DOW, air pressure
Lee et al. (2015)	2006–2008	Time-series	Shenyang	69.90	0.9963	0.9839	1.0090	Long-term and seasonal trends, temperature, relative humidity, DOW
Ma et al. (2011)	2006–2008	Case-crossover	Shenyang	75.00	1.0097	1.0001	1.0194	Long-term and seasonal trends, temperature, relative humidity, DOW, personal characteristics
Chen et al. (2018)	2013–2016	Time-series	Shijiazhuang	113.86	1.0040	1.0003	1.0079	Long-term and seasonal trends, temperature, relative humidity, DOW
Feng et al. (2018)	2013–2015	Time-series	Shijiazhuang	118.80	1.0063	1.0007	1.0119	Long-term and seasonal trends, temperature, relative humidity, DOW, public holidays
Huang et al. (2012)	2004–2008	Time-series	Xi'an	176.70	1.0019	0.9980	1.0059	Long-term and seasonal trends, temperature, relative humidity, DOW
Qi et al. (2017)	2013–2015	Case-crossover	Yinchuan	51.50	1.0022	1.0003	1.0041	Long-term and seasonal trends, temperature, relative humidity, DOW, public holiday
Yu et al. (2019)	2015–2016	Time-series	Changzhou	50.98	1.0165	1.0018	1.0315	Long-term and seasonal trends, temperature, relative humidity, DOW
Zeng et al. (2017)	2013–2015	Time-series	Chengdu	78.00	1.0031	1.0010	1.0052	Long-term and seasonal trends, temperature, relative humidity, DOW
Zhou et al. (2018)	2015–2017	Time-series	Fuzhou	27.72	1.0089	1.0001	1.0185	Long-term and seasonal trends, temperature, relative humidity, DOW
Yang et al. (2012)	2007–2008	Case-crossover	Guangzhou	70.10	1.0097	1.0016	1.0179	Long-term and seasonal trends, temperature, relative humidity, DOW ⁸
Lin et al. (2016b)	2012–2015	Time-series	Guangzhou	47.70	1.0106	1.0019	1.0194	Long-term and seasonal trends, temperature, relative humidity, DOW
Feng et al. (2015)	2013–2014	Time-series	Guangzhou	49.50	1.0160	0.9930	1.0380	Long-term and seasonal trends, temperature, relative humidity, DOW, public holiday
Xiong (2016)	2013–2015	Time-series	Guangzhou	46.83	1.0160	1.0072	1.0249	Long-term and seasonal trends, temperature, relative humidity, DOW
Lei et al. (2019)	2013–2015	Time-series	Hefei	77.79	1.0124	1.0029	1.0220	Long-term and seasonal trends, temperature, relative humidity, DOW, air pressure

Table 1 (continued)

Articles	Study period	Study design	City	Annual mean concentration of PM _{2.5} (µg/m ³)	RR or OR and 95% CI ⁵			Adjusted confounding factors
					RR	LCI ⁶	UCI ⁷	
Lin et al. (2016c)	1998–2011	Time-series	Hong Kong	37.50	1.0061	1.0019	1.0103	Long-term and seasonal trends, temperature, relative humidity, DOW, gaseous air pollutants, public holiday, influenza
He et al. (2016)	2011–2013	Time-series	Ningbo	60.10	1.0099	1.0019	1.0179	Long-term and seasonal trends, temperature, relative humidity, DOW, air pressure and wind velocity
Li et al. (2018a)	2014–2015	Time-series	Ningbo	45.20	1.0080	0.9896	1.0269	Long-term and seasonal trends, temperature, relative humidity, DOW, public holiday
Geng et al. (2013)	2007–2008	Time-series	Shanghai	53.90	1.0007	0.9871	1.0138	Long-term and seasonal trends, temperature, relative humidity, DOW
Qin et al. (2018)	2013–2016	Time-series	Shanghai	49.25	1.0080	0.9841	1.0319	Long-term and seasonal trends, temperature, relative humidity, DOW
Kan et al. (2007)	2004–2005	Time-series	Shanghai	56.40	1.0095	1.0016	1.0173	Long-term and seasonal trends, temperature, relative humidity, DOW
Cai et al. (2019)	2013–2015	Time-series	Shenzhen	35.00	1.0304	1.0060	1.0555	Long-term and seasonal trends, temperature, relative humidity, DOW, public holiday
Ge et al. (2015)	2010–2013	Time-series	Suzhou	57.40	1.0051	1.0010	1.0092	Long-term and seasonal trends, temperature, relative humidity, DOW
Zhu et al. (2017)	2012–2015	Time-series	Wuxi	66.00	1.0036	0.9989	1.0082	Long-term and seasonal trends, temperature, relative humidity, DOW
Wu et al. (2017)	2013–2015	Time-series	Zhuhai	34.23	1.0323	1.0015	1.0640	Long-term and seasonal trends, temperature, relative humidity, DOW, public holiday

¹Relative risk²Odds ratio³Respiratory disease⁴Ambient fine particulate matter with diameter less than 2.5 µm⁵Confidence interval⁶Lower confidence interval⁷Upper confidence interval⁸Day of week

other air pollutants and other health outcomes; (2) reviews and duplicate publications; (3) conference abstracts without sufficient information, government reports, and other non-peer-reviewed articles; (4) study duration of less than 1 year; (5) studies with inconvertible data; (6) studies without a city-specific quantitative short-term exposure–response relationship or only with stratified results; (7) studies only investigating specific high-risk subjects (e.g., infants, children, the elderly, pregnant women, patients, and smokers); (8) indoor, thermal, seasonal, occupational, or accidental exposures; (9) research on non-respiratory

diseases and single or sub-classification of respiratory diseases.

Short-term quantitative associations between air pollution exposure and health can be studied over a number of days (lags) (Atkinson et al. 2014). Lag is a commonly used concept in time-series analysis, which is used to predict future health status of the population through regression analysis performed using previous days' meteorological conditions or air pollution and current health conditions of people (Dai et al. 2004). The choice of lag length is a source of heterogeneity and may contribute to publication

Table 2 RRs¹ or ORs² of RD³ hospital outpatient visits due to short-term PM_{2.5}⁴ exposure

Articles	Study period	Study design	City	Annual mean concentration of PM _{2.5} (µg/m ³)	RR or OR and 95% CI ⁵			Adjusted confounding factors
					RR	LCI ⁶	UCI ⁷	
Lu et al. (2017)	2014–2015	Case-crossover	Beijing	87.40	1.0066	1.0058	1.0075	Long-term and seasonal trends, temperature, relative humidity, DOW ⁸ , air pressure
Duan et al. (2015)	2013	Time-series	Chengdu	96.90	1.0057	1.0048	1.0066	Long-term and seasonal trends, temperature, relative humidity, DOW
Guan et al. (2018)	2017	Time-series	Foshan	33.20	1.0274	1.0070	1.0483	Long-term and seasonal trends, temperature, relative humidity, DOW, air pressure, wind velocity
Jiang et al. (2017)	2015	Time-series	Guangzhou	38.60	1.0285	1.0131	1.0441	Long-term and seasonal trends, temperature, relative humidity, DOW
Ma et al. (2012)	2009–2011	Time-series	Guangzhou	72.00	1.0035	1.0012	1.0164	Long-term and seasonal trends, temperature, relative humidity, DOW, public holiday
Zheng et al. (2017)	2014–2015	Time-series	Guiyang	41.36	1.0064	1.0032	1.0091	Long-term and seasonal trends, temperature, relative humidity, DOW, air pressure, wind velocity
Li et al. (2018d)	2013–2016	Time-series	Hangzhou	62.25	1.0076	1.0041	1.0114	Long-term and seasonal trends, temperature, relative humidity, DOW
Liu et al. (2017)	2015	Time-series	Harbin	62.51	1.0052	1.0027	1.0076	Long-term and seasonal trends, temperature, relative humidity, DOW, public holiday
Zhang et al. (2015)	2013–2014	Time-series	Huai'an	73.70	1.0063	1.0037	1.0089	Long-term and seasonal trends, temperature, relative humidity, DOW
Zhao et al. (2017b)	2013–2015	Time-series	Jinan	110.80	1.0053	1.0042	1.0067	Long-term and seasonal trends, temperature, relative humidity, DOW, air pressure, wind velocity, precipitation
Wu (2018)	2015	Time-series	Linyi	77.45	1.0057	1.0041	1.0073	Long-term and seasonal trends, temperature, relative humidity, DOW, air pressure
Yang et al. (2017)	2013–2015	Time-series	Shanghai	55.50	1.0073	1.0059	1.0087	Long-term and seasonal trends, temperature, relative humidity, DOW
Hu et al. (2018)	2014–2016	Time-series	Shenzhen	29.86	1.0152	1.0062	1.0242	Long-term and seasonal trends, temperature, relative humidity, DOW
Peng et al. (2016)	2013	Time-series	Shenzhen	40.20	1.0181	1.0171	1.0191	Long-term and seasonal trends, temperature, relative humidity, DOW
Ma et al. (2017)	2015	Case-crossover	Taiyuan	60.11	1.0091	1.0044	1.0138	Long-term and seasonal trends, temperature, relative humidity, DOW
Zhang and Dai (2017)	2014–2015	Time-series	Zhoushan	31.30	1.0234	1.0117	1.0352	Long-term and seasonal trends, temperature, relative humidity, DOW

¹Relative risk²Odds ratio³Respiratory disease⁴Ambient fine particulate matter with diameter less than 2.5 µm⁵Confidence interval⁶Lower confidence interval⁷Upper confidence interval⁸Day of week

bias (Anderson et al. 2005). No universal standard of lags selection is found in previous studies (Zheng et al. 2015). In the present study, we followed Atkinson et al. (2012) protocols: if only one lag estimate was presented (either because only one was analyzed or only one was reported in the paper), then this estimate was recorded in the analysis. If more than one lag measure was presented, then we selected one for meta-analysis according to the following algorithm: (1) the lag that the author focused on or stated a priori; (2) the lag that was the most statistically significant (positive or negative), and (3) the lag with the largest effect estimate (positive or negative). For consistency, the distinctions between the measures of RR and OR were ignored in the present study, because the OR mathematically approximates the RR when the absolute risk of RD mortality and RD hospital outpatient visits owing to PM_{2.5} exposures in population is low; therefore, we extracted both of them in synthesis for simplicity (Greenland 1987; Sun et al. 2017). Drawing on prior research (Lu et al. 2015), all risk estimates recorded in the present study were expressed as or converted to standardized increment in RD mortality and RD hospital outpatient visits owing to per 10 µg/m³ change in PM_{2.5} concentration using the following formula:

$$RR_{\text{(Standardized)}} = RR_{\text{(Original)}}^{\text{Increment(10)/Increment(Original)}}$$

$$ER(\%) = (RR - 1) \times 100\%$$

Finally, 46 eligible studies were included according to the inclusion and exclusion criteria in this meta-analysis (Fig. 1). The title, authors, publication year, study period, study design, city, annual mean concentration of PM_{2.5}, health risk estimates (RR or OR and 95% confidence interval), and adjusted confounding factors were extracted and entered into Microsoft Excel database (Version 2010, Microsoft, Redmond, WA, USA). Stata Software (Version 12.0, StataCorp., College Station, TX, USA) was used to conduct a meta-analysis.

2.3 Quality control and assessment

2.3.1 Quality control

Two independent reviewers (Jiyao Sun and Meng Wang) performed a screening procedure to assure the quality of each study. They screened the titles and abstracts of retrieved literature. The full texts were reviewed and selected based on the inclusion and exclusion criteria. Disagreement and inconsistency were settled by discussion. If the argument still existed, then another reviewer (Xiaona Yan) was consulted to determine the final decisions. Three independent reviewers (Jiyao Sun, Meng Wang and Xiaona Yan) performed data extraction for

comparisons. Disagreements were resolved by consulting the fourth reviewer (Jian Wang).

2.3.2 Quality assessment

We evaluated the quality of the included studies according to a scale from a previously published meta-analysis (Song et al. 2016), which was composed of RD diagnosis, air pollutant measurements and adjustment of confounders. We determined if the diagnosis of RD was coded according to ICD-9 460-519 or ICD-10 J00-J98. For air pollutant measurements, we considered the study at low risks of bias if the frequency of air pollutant measurements were performed daily; otherwise, high risks of bias would be recognized. For adjustment of confounders, long-term trends, seasonality, meteorological parameters (temperature, relative humidity, air pressure and wind velocity), day of week (DOW), public holidays, and influenza epidemics were recorded to assess the quality of the studies. In the present research, we considered that studies which adjusted at least four above confounders were at low risks of bias.

2.4 Meta-analysis

Generic inverse variance method was employed to pool individual effects estimates in this meta-analysis. Specifically, the inverse variance-weighted average method summarizes effect sizes from multiple independent studies by calculating the weighted mean of the effect sizes using the inverse variance of the individual studies as weights (Lee et al. 2016; Sun et al. 2017). First, heterogeneity was examined across individual studies using the I^2 test. If I^2 exceeded 50%, then heterogeneity was observed; a random-effect model would be used to synthesize individual RRs or ORs; otherwise, a fixed-effect model would be used (Higgins et al. 2003). Meta-regression and subgroup analysis were performed to identify the source of the heterogeneity. Moreover, publication bias was explored with Egger's test at a p level of 0.05 (Egger 1997). The trim-and-fill method (Duval and Tweedie 2015) was used to adjust asymmetry if publication bias was observed. Finally, sensitivity analyses were performed to evaluate the robustness of the pooled effect estimates. In the present study, a p value of less than 0.05 was considered statistically significant.

3 Results

A total of 3432 related epidemiological works of literature were originally retrieved by systematically searching in PubMed, Web of Science, CNKI and Wanfang databases using the search criteria above. A total of 46 articles were

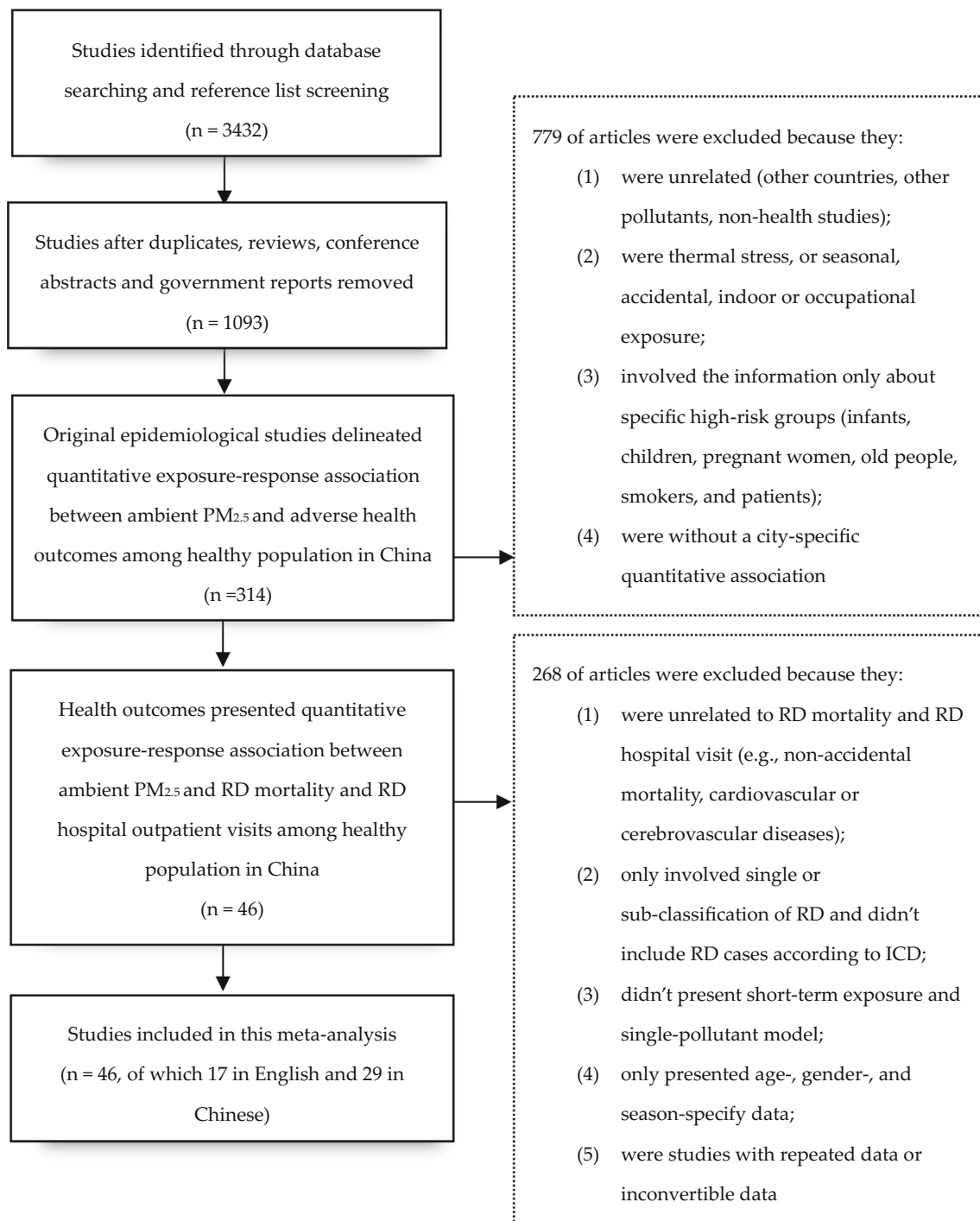


Fig. 1 Flow chart for study retrieval and selection process

finally selected in the present meta-analysis, including 29 Chinese studies and 17 English studies. These studies covered research across 27 cities, including 40 time-series studies and 6 case-crossover studies. RD mortality (Table 1) and RD hospital outpatient visits (Table 2) were reported by 30 studies and 16 studies, respectively. All included studies were at low risks of bias according to the

results of quality assessment. The annual mean concentrations of $\text{PM}_{2.5}$ ranged from 27.72 to 176.70 $\mu\text{g}/\text{m}^3$, and all reported concentrations exceeded the WHO AQGs (2006a) (Fig. 2). All RRs or ORs with their 95% confidence intervals in the following contexts of the present study were based on a 10 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ concentrations.

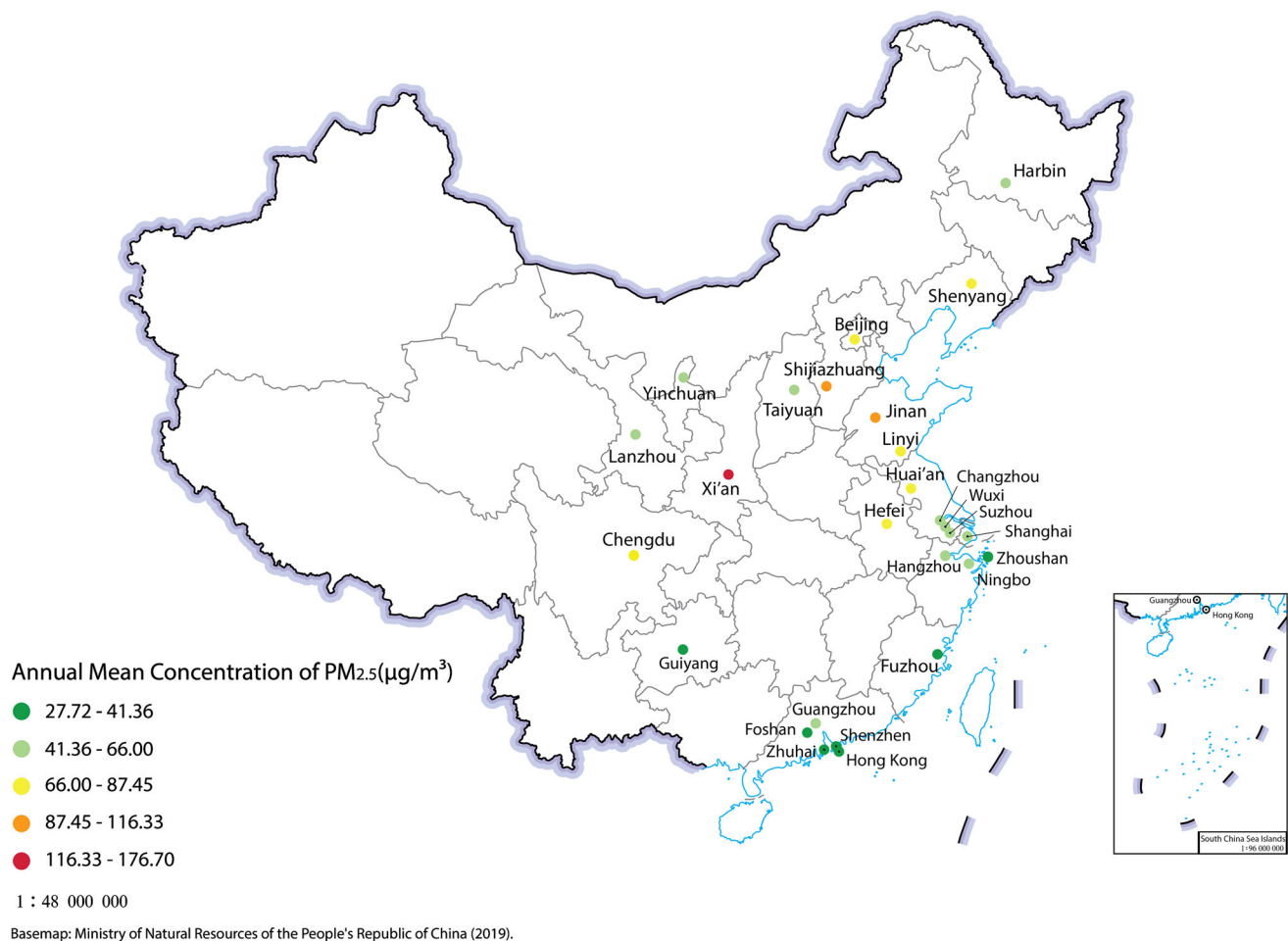


Fig. 2 Cities involved in this study and the annual mean concentration of PM_{2.5} of each city. (Note: If there are multiple articles reporting the PM_{2.5} concentrations in the same city, such as Guangzhou, we calculated the average values to represent the PM_{2.5} concentrations in this city)

The overall analyses of the present meta-analysis have demonstrated significant associations between ambient PM_{2.5} and RD in the Chinese population based on 46 studies. For RD mortality, combined estimate: 1.0048 (95% CI 1.0040, 1.0055); $I^2 = 38.4\%$; Egger's test: $p < 0.05$. For RD hospital outpatient visits, combined estimate: 1.0090 (95% CI 1.0062, 1.0118); $I^2 = 96.9\%$; Egger's test: $p > 0.1$.

3.1 RD mortality

Table 1 shows 30 epidemiological studies expressing the quantitative association between RD mortality and the change of a unit concentration of PM_{2.5} in short-term exposure. For each 10 μg/m³ increase in PM_{2.5} concentrations, the risk of RD mortality increased by 0.48% (95% CI 0.40%, 0.55%) (Fig. 3). No significant heterogeneity was observed ($I^2 = 38.4\%$).

3.1.1 Meta-regression

For all included studies, we conducted meta-regressions by geographical regions (south and north), annual mean concentrations of PM_{2.5}, and study designs (time-series and case-crossover). Results showed that geographical regions, annual mean concentrations of PM_{2.5}, and study designs did not exert a significant impact on the association between PM_{2.5} and the RD mortality ($p > 0.05$) (Table 3). We further illustrated the exposure–response relationship between PM_{2.5} and RD mortality. Result indicated that no significant linear association was identified between PM_{2.5} concentration and mortality due to RD (Fig. 4).

3.1.2 Publication bias and sensitivity analysis

Egger's test showed a significant existence of publication bias ($p < 0.05$). The trim-and-fill method was used to adjust asymmetry, which did not alter the direction of the combined effect (adjusted overall combined estimates:

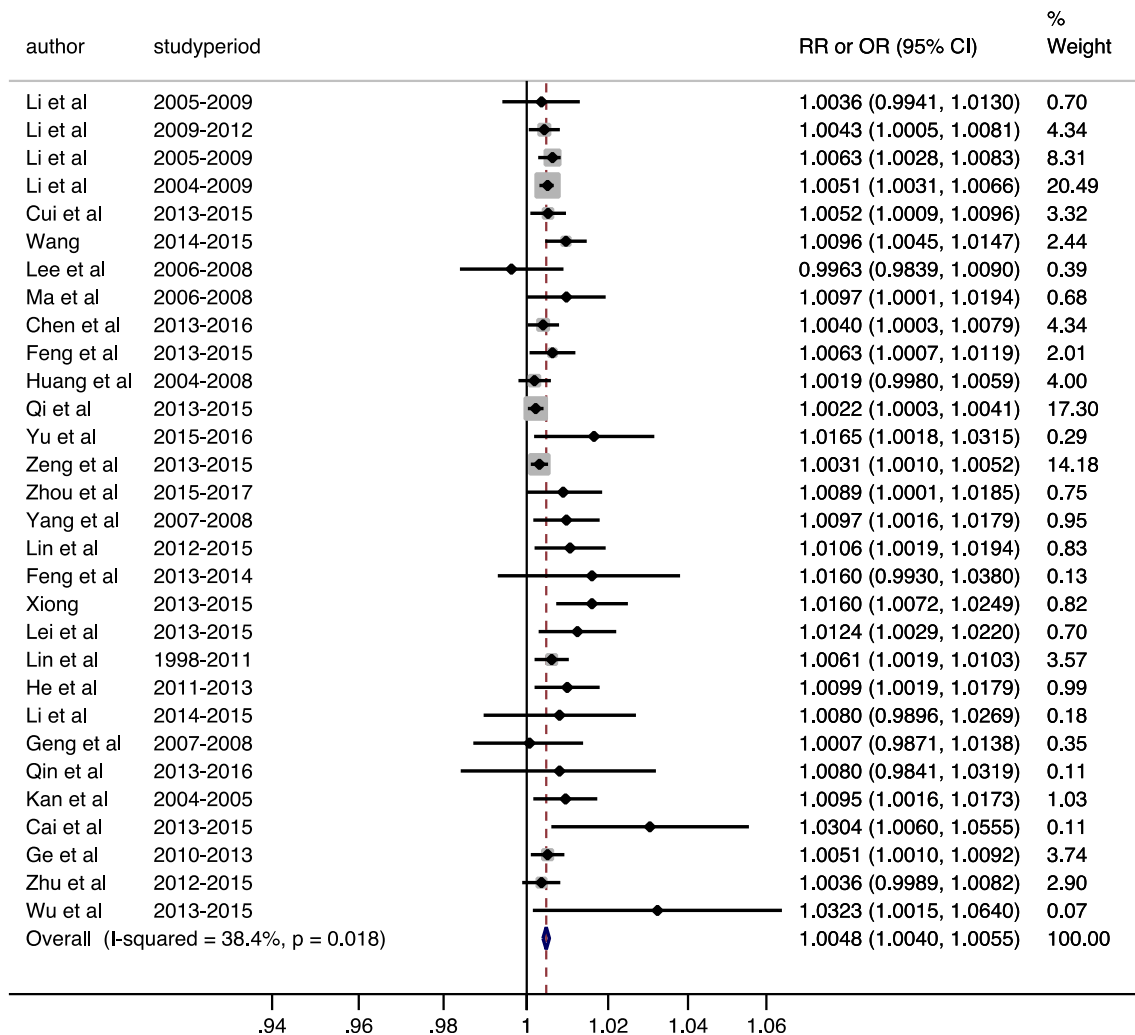


Fig. 3 The combined effect of RD mortality due to ambient PM_{2.5} short-term exposures

1.0050, 95% CI 1.0030, 1.0060). This result may suggest that the publication bias did not change the current positive association between PM_{2.5} and RD mortality.

Leave-one-out sensitivity analysis was performed to evaluate the robustness of the association results. The pooled estimate of the RD mortality remained stable after the sensitivity analysis, suggesting that the main results were not driven by any single study.

3.2 RD hospital outpatient visits

RD hospital outpatient visits owing to short-term exposure PM_{2.5} were reported in 16 studies, spanning across 14 cities in China. Table 2 presents detailed information on each study. For each 10 µg/m³ increase in PM_{2.5} concentrations, the pooled effect estimate of RD hospital outpatient visits was 1.0090 (95% CI 1.0062, 1.0118). Significant heterogeneities were identified across studies ($I^2 = 96.9\%$) (Fig. 5).

Table 3 Results of meta-regressions

Category	Regression coefficients	SE	T	P	95% CI	
					LCI	UCI
Region	1.0023000	0.0013184	1.75	0.092	0.9996034	1.005005
Concentration	0.9999614	0.0000200	− 1.93	0.064	0.9999204	1.000002
Study Design	0.9981781	0.0015241	− 1.19	0.242	0.9950610	1.001305

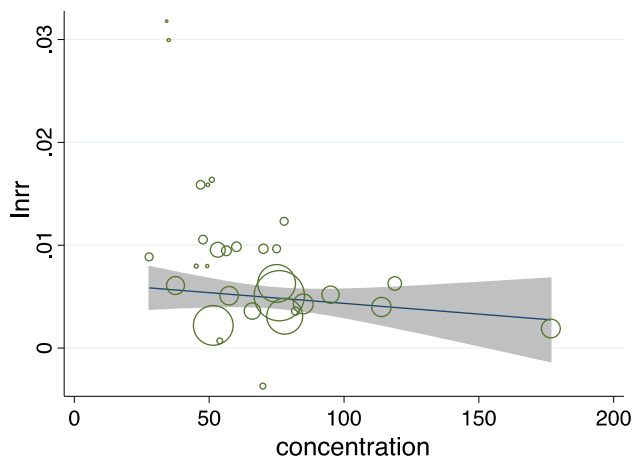


Fig. 4 Meta-regression analysis of annual mean $PM_{2.5}$ concentration and RD mortality. (Note: Each open circle represents a study. The size of the circle indicates the precision of the effect estimate and the weight given to that study.)

3.2.1 Meta-regression

Meta-regressions were performed to identify the source of heterogeneity by stratifying the studies on the basis of geographical regions (south and north), annual mean concentrations of $PM_{2.5}$, and study designs (time-series and

case-crossover). Results show that only the annual mean concentration of $PM_{2.5}$ had a significant impact on the association between $PM_{2.5}$ and RD hospital outpatient visits ($p < 0.05$) (Table 4). A downward trend in the increased risks of RD hospital outpatient visits was found with an increase in annual mean $PM_{2.5}$ concentrations (Fig. 6).

3.2.2 Subgroup analysis

On the basis of the result of the meta-regression analysis, a subgroup analysis was conducted to explore further the associations between RD hospital outpatient visits and $PM_{2.5}$ concentrations. In the process of grouping, we first sorted the articles according to the $PM_{2.5}$ concentrations from low to high. The articles with higher $PM_{2.5}$ concentrations were then accumulatively combined with those having lower concentrations one by one. Over several simulations, each simulation divided the articles into two groups: the combined articles (group 1) and the remaining articles excluded from the combination (group 2). For example, in the first simulation, the combination of those article with $PM_{2.5}$ concentrations of $31.30 \mu\text{g}/\text{m}^3$ and $29.86 \mu\text{g}/\text{m}^3$ constituted group 1, and the remaining articles constituted group 2. Then subgroup analysis was conducted

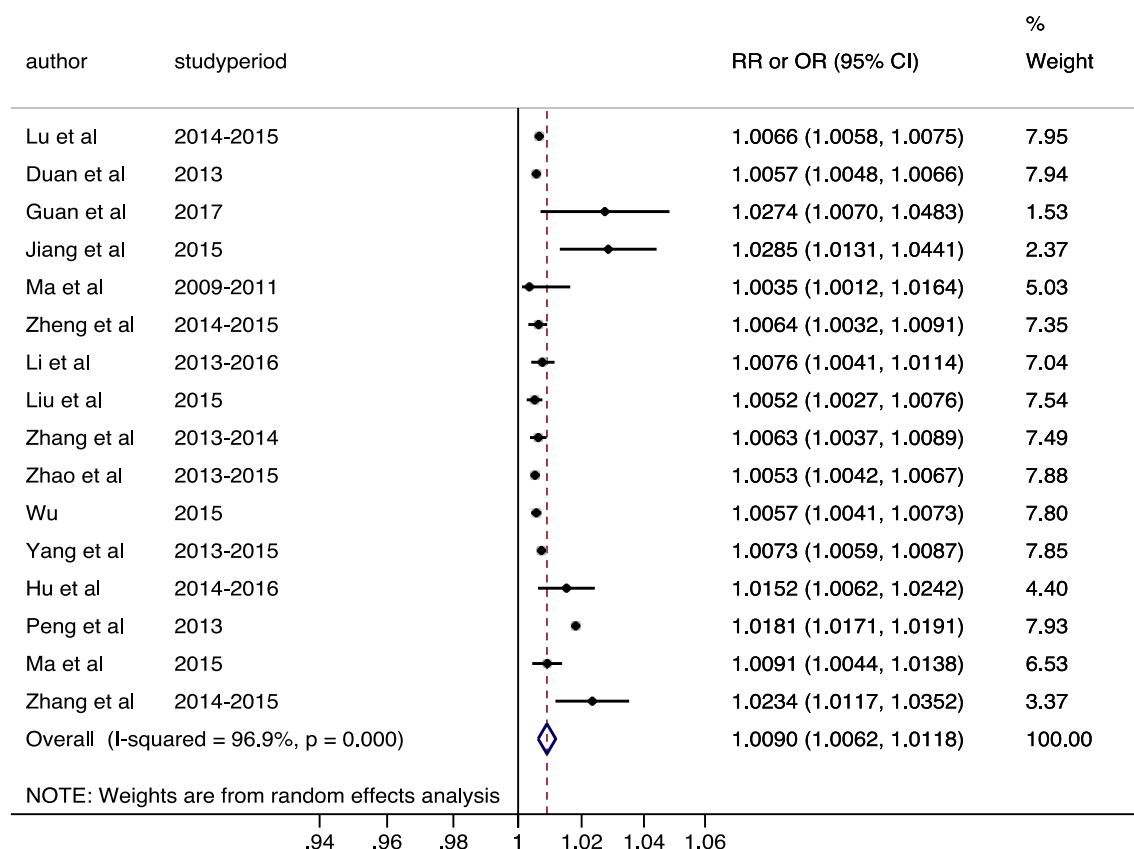
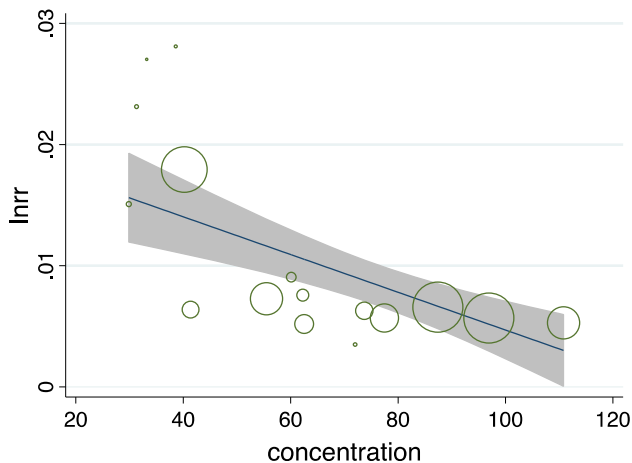


Fig. 5 The combined effects of RD hospital outpatient visit due to $PM_{2.5}$ short-term exposures

Table 4 Results of meta-regression

Category	Regression coefficients	SE	T	P	95% CI	
					LCI	UCI
Region	1.0043110	0.0029009	1.49	0.159	0.9981086	1.0105520
Concentration	0.9998535	0.0000476	− 3.08	0.008	0.9997514	0.9999555
Study Design	0.9984374	0.0043973	− 0.36	0.728	0.9890505	1.0079130

**Fig. 6** Meta-regression analysis of annual mean PM_{2.5} concentration and RD outpatient visits. (Note: Each open circle represents a study. The size of the circle indicates the precision of the effect estimate and the weight given to that study.)

to test if heterogeneity reduced in both groups. Subsequently, in the second simulation, group 1 involved articles with PM_{2.5} concentrations of 33.20 µg/m³, 31.30 µg/m³, and 29.86 µg/m³, and the group 2 was composed of the remaining articles, then same analytical procedure was carried out. The procedure of grouping continued until heterogeneity in both groups significantly decreased. Finally, results showed that the heterogeneity disappeared in both groups after stratifying the models by levels of PM_{2.5} concentrations in the range of 29.86–40.20 µg/m³ and 41.36–110.80 µg/m³. In the range of 29.86–40.20 µg/m³ of annual mean PM_{2.5} concentrations, each 10 µg/m³ increase in PM_{2.5} concentrations was associated with an increased risk of RD hospital outpatient visits of 1.82% (95% CI 1.72%, 1.92%), which was higher than the overall combined risk of 0.90%. However, in the range of 41.36–110.80 µg/m³, the pooled estimate of RD hospital outpatient visit reduced to 0.62% (95% CI 0.57%, 0.66%) with each additional 10 µg/m³ increase in PM_{2.5} concentrations, which was lower than the overall risk. The increased risk of RD hospital outpatient visits (0.62%) owing to short-term PM_{2.5} exposures was lower in the higher PM_{2.5} concentration range than that (1.82%) in the lower range (Fig. 7).

3.2.3 Publication bias and sensitivity analysis

Egger's test showed that no significant publication bias effect was detected ($p = 0.927 > 0.1$). The result of the sensitivity analysis illustrated that the combined RRs or ORs remained stable, indicating that the results of the present study were not driven by any single study.

4 Discussion

In this systematic review and meta-analysis, we assessed the quantitative associations between short-term PM_{2.5} exposures and RD. Results demonstrated that each 10 µg/m³ increase in PM_{2.5} concentration corresponded to 0.48% (95% CI 0.40%, 0.55%) increase in RD mortality and 0.90% (95% CI 0.62%, 1.18%) increase in RD hospital outpatient visits respectively in the Chinese population. A downward trend in the ER of RD hospital outpatient visits was observed with an increase in annual mean PM_{2.5} concentrations. After stratifying by PM_{2.5} concentrations, we found the combined risk of RD hospital outpatient visits (0.62%) owing to short-term PM_{2.5} exposures was lower in the higher PM_{2.5} concentration range than that (1.82%) in the lower range with per 10 µg/m³ increase in PM_{2.5} concentrations.

The overall magnitudes of acute estimates of exposure to PM_{2.5} on RD found in the present research could be compared with those identified in other studies. Chen (2013) reported a combined estimate of 0.46% increase in RD mortality per 10 µg/m³ increase in PM_{2.5} in 17 Chinese cities, which was highly similar to our estimate. Shang et al. (2013) and Liu et al. (2015) systematic reviews and meta-analysis studies reported summarized estimates of 0.51% and 0.50% increases in RD mortality, respectively, per 10 µg/m³ increases in PM_{2.5} concentrations in the Chinese population, showing similar magnitudes with our finding. Newell et al. (2017) conducted a meta-analysis by examining 85 studies across 12 LMICs and observed a 10 µg/m³ increase in same-day PM_{2.5} was associated with a 0.57% increase in RD mortality, which approximated our result in the present study. Zhong et al. (2017) meta-analysis also indicated a roughly similar increased risk of RD mortality (0.56%) owing to per 10 µg/m³ increase in PM_{2.5} concentration in the Chinese population.

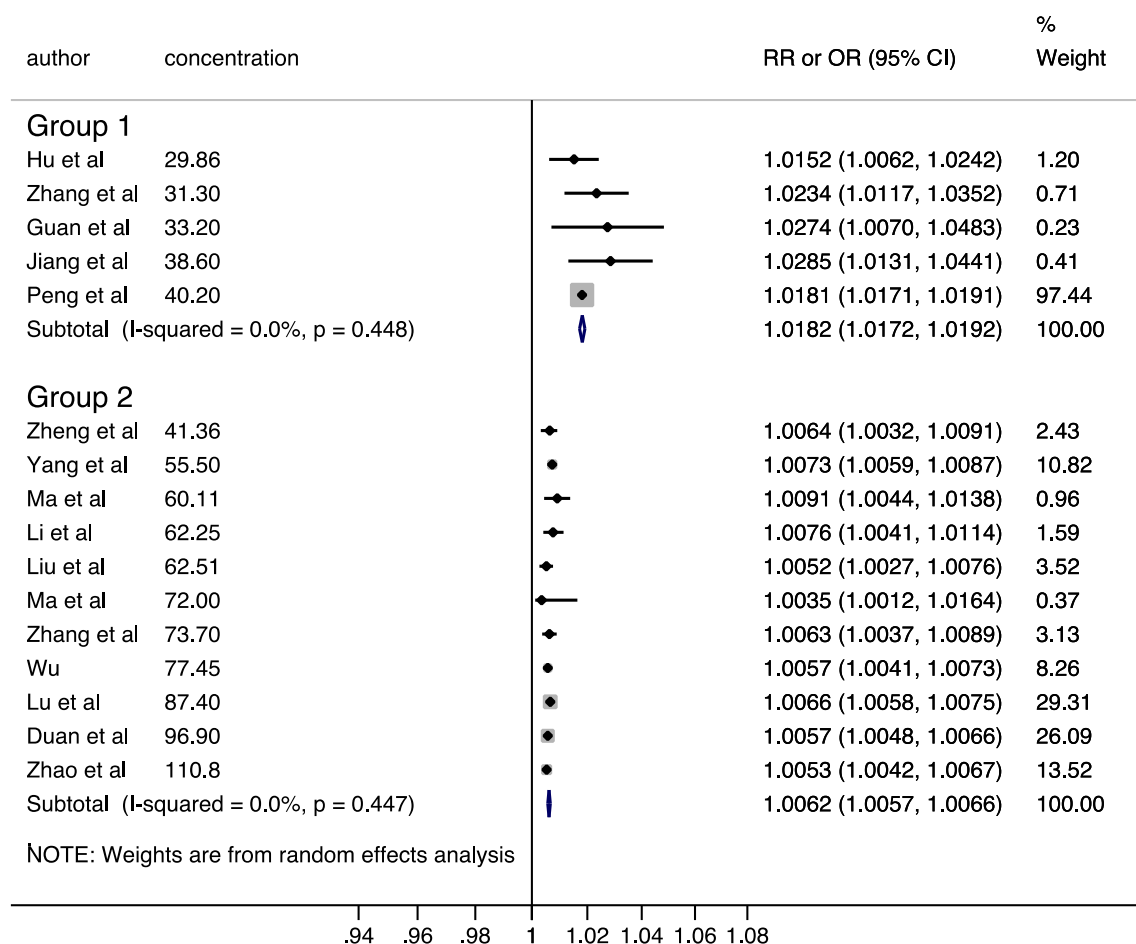


Fig. 7 Subgroup analysis of annual mean PM_{2.5} concentration and RD outpatient visits

Moreover, some multi-city or meta-analysis studies reported relatively higher or lower quantitative associations between PM_{2.5} and RD mortality in China, compared with our result drawn from the present analysis. For example, Lu's (2015) systematic review and meta-analysis illustrated that a 10 µg/m³ increase in PM_{2.5} was associated with a 0.75% increase in RD mortality. Lin et al. (2016a) found that short-term exposure to outdoor PM_{2.5} was associated with an increased risk of 1.68% in RD mortality by conducting a multi-city study in the Pearl River Delta in China. Lee et al. (2015) analysis in East Asia, including six Chinese cities, Taipei, Taichung, Kaohsiung, Beijing, Guangzhou and Shenyang revealed a 1% increase in RD mortality owing to each 10 µg/m³ increase in PM_{2.5} concentration. However, a lower magnitude of RD mortality was reported in Wang et al. (2019) multi-city study, which demonstrated that per 10 µg/m³ increase in PM_{2.5} was associated with an increase of 0.29% of RD mortality. A similar situation also occurred in Ren et al. (2017) meta-analysis that for single-day lags, per 10 g/m³ increase in PM_{2.5} concentration was associated with a 0.30% increase

in daily respiratory deaths. The discrepancy of the reported results across studies may be attributed to the changes among Chinese cities in levels of exposure; chemical compositions of ambient air pollution; city characteristics, such as urbanization level, medical and hygiene standards, and age structures; and the underlying susceptibility of the population during the rapid economic development in recent years (Lai et al. 2013). The chemical composition of PM_{2.5} pollutant varies in different cities across China due to distinctions in energy consumption structures, economic development, terrain and meteorological conditions (Cao 2016). For example, several highly developed cities, such as Beijing, Tianjin and Guangzhou, reported higher level of organic carbon (OC) and elemental carbon (EC) in PM_{2.5} chemical components than lesser-developed cities, such as Lin'an in Zhejiang Province in recent years (Zhao et al. 2014). Cao and colleagues (2012) demonstrated significant positive associations of RD mortality with OC and EC. Also, a meta-analysis of short-term exposure time-series studies by Smith et al. (2009) consistently reported that EC was positively associated with cardiopulmonary mortality.

Besides, the variations of lag selection across studies may also introduce heterogeneity by exaggerating or dampening the effect value (Anderson et al. 2005).

Compared with the quantitative dose–response association between $PM_{2.5}$ and RD mortality found in the present study, the combined estimates reported in developed countries with low levels of $PM_{2.5}$ pollution appeared to be high. For example, Zanobetti and Schwartz (2009) conducted a study including 112 U.S. cities and reported a 1.68% increase in RD mortality for a $10 \mu g/m^3$ increase in two-day averaged $PM_{2.5}$. Dai et al. (2014) and Franklin et al. (2007) multicity analysis in 75 and 27 U.S. cities, respectively, estimated a similar 1.71% and 1.78% increase in RD deaths in the association with a $10 \mu g/m^3$ increase in $PM_{2.5}$ concentration. Ostro et al. (2006) study in nine California counties also illustrated a 2.2% increase in RD mortality with a $10 \mu g/m^3$ change in two-day average $PM_{2.5}$ concentration. A similar case in another multi-city study including 10 European–Mediterranean metropolitan areas uncovered that a $10 \mu g/m^3$ increase in $PM_{2.5}$ gave rise to a 1.91% increase in RD mortality (Samoli et al. 2013). Achilleos et al. (2017) examined the same association in the U.S., South America, Canada, Europe, and Western Pacific through a meta-analysis, reporting a 1.10% increase in RD mortality per $10 \mu g/m^3$ increase in $PM_{2.5}$. The disparities between China and these developed areas may have been because of the differences in population size, severities, the sources of air pollution, and chemical compositions of PM in various study locations (Shang et al. 2013).

Cui et al. (2013) epidemiological study on the hospital outpatient visits due to RD in Harbin, China, from 2009 to 2011 showed that when $PM_{2.5}$ concentration increased by 44, 35 and $60 \mu g/m^3$, the number of RD outpatient visits increased by 11.6%, 18.9% and 35.8%, respectively. In the present study, we observed that an increase of $10 \mu g/m^3$ of $PM_{2.5}$ concentrations was associated with a 0.90% increase in RD hospital outpatient visits in the Chinese population. However, our result was slightly higher than the result reported in Liang et al. (2016) meta-analysis, which demonstrated an increase of 0.5% of the RD hospital outpatient visit per $10 \mu g/m^3$ increase in $PM_{2.5}$ concentrations. The slightly stronger effect estimate in our study may be attributable to the distinct number of included articles. Special attention should be paid to the result from meta-regression that a downward trend in the ER of RD hospital outpatient visits with an increase in annual mean $PM_{2.5}$ concentrations was found. Subgroup analysis also verified that the combined effect of RD hospital outpatient visits owing to short-term $PM_{2.5}$ exposures was lower in the higher $PM_{2.5}$ concentration range than that in the lower range per $10 \mu g/m^3$ increase in $PM_{2.5}$ concentrations. After the database research, no previous study has reached such a

conclusion based on the pollutant–outcome pair of $PM_{2.5}$ and RD hospital outpatient visits in other LMICs that may have comparable air pollution level to China; however, Lu et al. (2015) meta-analysis identified a similar declining trend in the association of PM_{10} and RD mortality, which could offer reference to our results in the present study. Daniels et al. (2000) study in the U.S. and Samoli et al. (2005) research in Europe identified a linear dose–response association between PM pollution and mortality; that is, the incremental effect of an increase in PM on mortality is the same for all PM exposures. However, they did not reveal the quantitative exposure–response relationship among $PM_{2.5}$, morbidity, and hospital outpatient visits. The association was identified in much less polluted developed areas, compared with the present study in the Chinese context. China, as the biggest developing country, has undergone rapid economic growth through industrialization and urbanization, characterized by increased building, manufacturing, and traffic density, leading to higher concentrations of air pollutants (Lee et al. 2015). Katsouyanni et al. (1997) mentioned that the concentration–response of health outcomes to air pollutants is dampened at higher concentrations, and Lu et al. (2015) indicated that the shape of dose–response curves tends to be highly stable and represented a departure from the linear dose–response relation when reaching a certain high PM concentration. Ambrose and Barua's (2004) research, from the pathophysiological perspective, discussed the plausibility of a nonlinear response relationship between cardiovascular health and cigarette smoking, suggesting that the identified nonlinear response may be caused by underlying biochemical and cellular processes becoming saturated with small doses of toxic components. Luo et al. (2016) suggested living in areas with high pollution concentrations may increase people's disease tolerance and reduce susceptibility to respiratory diseases. Although the above statement contended the possibility, the exact mechanisms of such phenomenon involved in the present study call for further investigation.

Several limitations of the present study should be considered. First, publication bias was detected in the pollutant–outcome pair of $PM_{2.5}$ and RD mortality. This bias may be because the authors are more likely to emphasize and publish positive results than negative ones (Sun et al. 2017). Second, time-series studies and case-crossover studies were included in the present study, which may affect the individual effects and lead to heterogeneities across studies (Jaakkola 2003). Third, a small number of fixed outdoor monitors as the standard surrogate for personal exposure throughout the included studies were less likely to represent the true personal experiences of exposure (Newell et al. 2017). Most included articles collected the data of RD outpatient visits only from one or two

municipal public hospitals, which could not fully represent the actual morbidity of respiratory diseases in the city. Fourth, temperature-, season-, age-, and gender-specific data were not collected and performed in the present study, which could also provide valuable information on the adverse health effects of ambient PM_{2.5}. Moreover, only the single-pollutant model results were included. We did not consider the potential synergistic effects from other pollutants. Finally, although a declining trend was found in the ER of RD hospital outpatient visits with an increase in annual mean PM_{2.5} concentrations, our analysis may still lack a sufficient number of epidemiological studies to uncover the real dose–response relationship (Roberts and Martin 2006). Therefore, further research needs to be conducted to add convincing evidence by analyzing studies that are relevant.

5 Conclusions

The present meta-analysis provides updated scientific evidence that short-term exposures to ambient PM_{2.5} were associated with increased risks of RD mortality and RD hospital outpatient visits in the Chinese population. We illustrated a downward trend in the ER of RD hospital outpatient visits with an increase in annual mean PM_{2.5} concentrations. Evidence presented in this study could help guide Chinese policymakers to improve air quality and alleviate the adverse health impact of PM_{2.5} pollutions.

Acknowledgements The authors would like to express gratitude to Junyan Ye for helping to make the map. We also thank Yingjie Fu, Huicui Shan and Linlin Yin for providing necessary materials and helpful comments on the article.

Author contributions JS and JW conceived and designed the research. JS, XY and MW retrieved and screened literatures and extracted data. JS conducted quantitative meta-analysis and finished the first version of the manuscript. NZ and JW provided project supervision and guidance to the writing of the article. They also reviewed, edited and modified the manuscript critically for some important content.

Funding This research was funded by the Social Science Planning and Research Project of Shandong Province (19CZKJ02) and the Health Industry Research Project on Medical and Health Care of Shandong University (19SDUY01). Nan Zhang was partly supported by Cathie Marsh Institute for Social Research seedcorn funding about the impact of air pollution on wellbeing of older people in the Global South, and Global Challenge Research Fund (GCRF) Pump-Priming Grant.

Availability of data and materials All data analyzed during this study are included in this published article.

Compliance with ethical standards

Conflict of interest The authors declared no conflict of interest.

Ethical approval Data were collected from published literatures rather than human subjects. No personal private information was included.

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